Yating Zou

Department of Biostatistics
Gillings School of Public Health
University of North Carolina at Chapel Hill
UNC Gillings Center for Artificial Intelligence and Public Health (CAIPH)

Personal Website
Google Scholar
github.com/yatingz205

RESEARCH AREAS

My research develops nonparametric methodology at the interface of distributional estimation, kernel methods, and reinforcement learning, with a focus on supporting sequential decision processes for real-world systems in structured environments, including dependent, missing, and hierarchical data, complex survey designs and adaptive trials.

EDUCATION

† *Indicates expected*

2023–2027 † Ph.D. in Biostatistics, University of North Carolina at Chapel Hill
Dissertation: "From Design to Decision: Distributional Estimation and Sequential Decision Optimization under Structured Environments"
Advisor: Michael R. Kosorok;
Other Committee Members: Michael H. Hudgens, Didong Li, Ivana Malenica, Paul Lanier

2019–2022 B.S.P.H. in Biostatistics, University of North Carolina at Chapel Hill
Honors Thesis: "A Review of Causal Average Treatment Effect and Treatment Heterogeneity Estimation with Non-finite Action"
Thesis Advisor: Michael R. Kosorok;
Other Committee Members: Donglin Zeng, Jane Monaco

2018–2022 B.S. in Mathematics, University of North Carolina at Chapel Hill

PUBLICATIONS

† *First or joint-first author with equal contribution*
‡ *Joined as senior statistician*

Methodology
Submitted / preprint

[1] **Y. Zou**†, H. Hu, G. Kremling, and J. Näf. *Asymptotics of Nonparametric Estimation under General Non-monotone M(N)AR Missingness: A Nonparametric Maximum Likelihood Approach*. 2026. DOI: 10.48550/arXiv.2608.10113. arXiv: 2608.10113 [stat.ME].

[2] **Y. Zou**†, M. Matabuena, and M. R. Kosorok. *Distributional Random Forests for Complex Survey Designs*. Under review. Dec. 2025. DOI: 10.48550/arXiv.2512.08179. arXiv: 2512.08179 [stat.ME].

[3] J. P. Zitovsky†, **Y. Zou**†, L. Wilson, and M. R. Kosorok. *Latent Utility Q-Learning for Preference-Adaptive Dynamic Treatment Regimes*. In invited revision with JASA. July 2023. DOI: 10.48550/arXiv.2307.12022. arXiv: 2307.12022 [stat.ME].

[4] K. Lee, **Y. Zou**, and M. R. Kosorok. *What a Data Release Must Preserve for Off-Policy Evaluation*. Submitted to AISTATS 2027. Oct. 2026. DOI: 10.48550/arXiv.2610.15283. arXiv: 2610.15283 [stat.ML].

[5] Z. A. A. Hammouri, **Y. Zou**, R. Ghosal, J. C. Vidal, and M. Matabuena. *ROC Analysis with Covariate Adjustment Using Neural Network Models: Evaluating the Role of Age in the Physical Activity–Mortality Association*. Oct. 2025. DOI: 10.48550/arXiv.2510.14494. arXiv: 2510.14494 [stat.ME].

[6] N. L. B. Freeman, **Y. Zou**, K. Lee, C. Yu, D. Li, and M. R. Kosorok. *The Fidelity and Feedback Traps: The Case for Health Digital Twins as Modular Evolving Causal Systems*. 2026. DOI: 10.48550/arXiv.2607.27192. arXiv: 2607.27192 [stat.OT].

In preparation (available upon request)

[7] A. Wang, **Y. Zou**, and M. R. Kosorok. *Test-Set Validation of Dynamic Treatment Regimes for Indefinite-Horizon Survival Outcomes*. In preparation. 2026.

[8] M. Matabuena and **Y. Zou**. *Covariate-Adjusted ROC Analysis in Metric Spaces*. In preparation. 2026.

Applied & Collaborative

Published

[9] J. Gingles, **Y. Zou**‡, L. Chinula, J. H. Tang, I. Kaliati, T. Mkochi, L. Msowoya, C. Nakanga, F. Saidi, J. S. Smith, et al. "Persistent Human Papillomavirus After Thermal Ablation in Malawian Women With and Without HIV: Implications for Risk-Based Cervical Cancer Prevention in Low- and Middle-Income Countries". In: *Women's Health (London)* 22 (Apr. 2026). DOI: 10.1177/17455057261428547.

[10] N. E. Rosenberg, A. M. Young, **Y. Zou**, J. Justman, M. E. Charurat, S. Moyo, K. Zuma, L.-G. Bekker, L. Stranix-Chibanda, S. J. Phiri, et al. "An HIV-1 Risk Assessment Tool for Men Aged 15-59 years in 13 African Countries: A Pooled Analysis of Nationally Representative Surveys". In: *JAIDS Journal of Acquired Immune Deficiency Syndromes* 101.2 (Feb. 2026), pp. 173–182. DOI: 10.1097/QAI.0000000000003773.

[11] C. Mungo, G. K. Ellis, M. Rop, **Y. Zou**‡, J. Omoto, and L. Rahangdale. "Perceived acceptability of self–administered topical therapy for cervical precancer treatment among women undergoing cervical cancer screening in Kenya". In: *Advances in Global Health* 4.1 (Jan. 2025), p. 2329438. DOI: 10.1525/agh.2025.2329438.

[12] C. Mungo, K. Adewumi, G. Ellis, M. Rop, E. Adoyo, **Y. Zou**‡, and L. Rahangdale. "Men's Perceptions and Perceived Acceptability of Their Female Partner's Use of Self-Administered Intravaginal Therapies for Treatment of Cervical Precancer in Kenya". In: *ecancermedicalscience* 18 (Feb. 2024), p. 1719. DOI: 10.1101/2024.02.06.24302397.

[13] N. Rosenberg, B. Shook-Sa, A. Young, **Y. Zou**, L. Stranix-Chibanda, M. Yotebieng, N. Sam-Agudu, S. Phiri, W. Mutale, L. Bekker, et al. "A Human Immunodeficiency Virus Type 1 Risk Assessment Tool for Women Aged 15–49 Years in African Countries: A Pooled Analysis Across 15 Nationally Representative Surveys". In: *Clinical Infectious Diseases* 79.5 (Nov. 2024), pp. 1223–1232. DOI: 10.1093/cid/ciae211.

[14] K. Yang, J. Tian, B. Zhang, M. Li, W. Xie, **Y. Zou**, Q. Tan, L. Liu, J. Zhu, A. Shou, et al. "A multidimensional nomogram combining overall stage, dose volume histogram parameters and radiomics to predict progression-free survival in patients with locoregionally advanced nasopharyngeal carcinoma". In: *Oral Oncology* 98 (2019), pp. 85–91. DOI: 10.1016/j.oraloncology.2019.09.022.

Submitted / preprint

[15] M. M. Diaz, K. Enders, S. Tovar-Ramirez, Y. Rodriguez-Angeles, V. Roldan, M. Nolasco, **Y. Zou**‡, J. She, P. Sotolongo, F. Mejia, et al. "Development and Validation of Regression-based Neuropsychological Testing Norms for Peruvian adults to detect HIV-associated Neurocognitive Impairment". In: *medRxiv* (Feb. 2026). DOI: 10.64898/2026.02.09.26345550.

In preparation (available upon request)

[16] **Y. Zou**†‡, L. Pluta, M. Kang, R. F. Ambinder, S. E. Krown, T. B. Campbell, M. Epeldegui, M. Z. Borok, M. C. Hosseinipour, and D. P. Dittmer. *Kaposi Sarcoma Herpesvirus in Plasma: Association*

with Treatment Failure in Limited-Stage, Treatment-Naive AIDS-KS. Selected as oral in CROI 2026. June 2026.

[17] B. Rowland, G. Sowa, B. Zhao, A. Ivanova, S. Mackey, W. Marras, S. Rundell, Y. Shao, A. Wasan, **Y. Zou**, et al. *Results from the Biomarkers for Evaluating Spine Treatments (BEST) Trial: 24-Week Outcomes*. In preparation. 2026.

[18] B. Zhao, G. A. Sowa, A. Ivanova, K. M. Kidwell, M. R. Kosorok, …, **Y. Zou**, K. J. Anstrom, D. J. Clauw, and M. C. Mauck. *Identifying Response Heterogeneity in the BEST Trial of Chronic Low Back Pain*. 2026.

TEACHING EXPERIENCE

Weekly one-hour office hour + homework assignment marking
† *Two guest lectures*

2026 Fall	BIOS 791†: Empirical Processes and Semiparametric Inference	Ph.D. elective
2025 Fall	BIOS 777: Precision Medicine	Ph.D. elective
2024 Spring	BIOS 791: Empirical Processes and Semiparametric Inference	Ph.D. elective
2024 Spring	BIOS 500H: Introduction to Biostatistics	Undergraduate core
2023 Fall	BIOS 760: Advanced Probability and Statistical Inference I	Ph.D. core
2022 Spring	MATH 521: Advanced Calculus I (Real Analysis)	Undergraduate core

PRESENTATIONS

† *Forthcoming*

Invited / Contributed Talk

[1] "Asymptotics of Nonparametric Estimation under general non-monotone MAR missingness"
CFE-CMStatistics 2026, Berlin, Germany† (presenting by Jeffrey)

[2] "Test-Set Validation of Dynamic Treatment Regimes for Indefinite-Horizon Survival Outcomes"
JSM 2026, Boston, USA, (presented by Anthony Wang)
⋆ Received the Zink Scholarship in Biostatistics

[3] "Distributional Random Forest for Complex Survey Designs"
JSM 2025, Nashville, USA

[4] "Preference Integrated Dynamic Treatment Regimes: Methodology and Application to SMARTs"
CFE-CMStatistics 2025, London, UK

[5] "Kaposi Sarcoma Herpesvirus in Plasma: Association with Treatment Failure in Limited-Stage AIDS-KS"
CROI 2026 (Conference on Retroviruses and Opportunistic Infections), Colorado, USA

Poster Presentations

[1] "Preventing Youth Mental Health Emergencies in NC: the Dynamic Treatment Regime Approach"
Annual Research Meeting 2026, Seattle, USA
UNC Data Science Symposium 2025, North Carolina, USA

[2] "An HIV-1 risk assessment tool for women aged 15-49 in African countries: A machine-learning approach across 15 pooled nationally representative surveys"
CROI 2024 (Conference on Retroviruses and Opportunistic Infections), Colorado, USA
UNC Data Science Symposium 2024, North Carolina, USA

[3] "Kaposi Sarcoma Herpesvirus in Plasma: Association with Treatment Failure in Limited-Stage AIDS-KS"
UNC Lineberger 50th Annual Scientific Retreat 2026, North Carolina, USA

SOFTWARE AND TOOLS

A SMART Trial Analysis Pipeline: Estimation, Inference, and Interpretation
R | targets | Quarto | tidymodels
Targets-orchestrated R pipeline for the Biomarkers for Evaluating Spine Treatments (BEST) trial (Registration ID: NCT05396014), estimating dynamic treatment regimes via Q-learning and super-learner ensembles, with CV-TMLE inference, variable-importance and interpretable modeling, and reproducible reporting.

UNC Gillings Center for Artificial Intelligence and Public Health (CAIPH)
UNC Collaborative Studies Coordinating Center (CSCC)

LLM Empowered Validation Agent for Health Resource Tracking in North Carolina
Python | LangChain | RAG | Provider-agnostic LLM
A provider-agnostic, tool-using LLM agent that validates healthcare-location records for MAHEC's FindMyCareWNC platform. Automatically flagging out-of-date entries and detecting new locations, with a human-in-the-loop review interface for performance monitoring.

UNC Gillings Center for Artificial Intelligence and Public Health (CAIPH)
Mountain Area Health Education Center (MAHEC)

HONORS AND AWARDS

2026	Kalyani Sen International Student's Scholarship in Biostatistics
2025	Travel Award, Biostatistics Department at Gillings School of Public Health
2025	Gillings School Graduate Student Teaching Assistant Award
2023	Delta Omega Honor Society for Public Health, Theta Chapter
2023	Highest Distinction and Highest Honor, UNC-CH
2022	Salisbury Family Senior Thesis Research Award

PROFESSIONAL EXPERIENCE

2026	*Graduate Research Assistant* Gillings Faculty AI Workshop Series Focus: Supporting a Dean-funded AI training workshop series for Gillings faculty, in partnership with AWS (Amazon), spanning omics, agentic AI, multimodal data integration, and mHealth applications.
2025–Present	*Graduate Research Assistant* UNC Gillings Center for Artificial Intelligence and Public Health (CAIPH), Focus: Methodological and application research in machine learning and precision medicine.
2022–2025	*Graduate Research Assistant* UNC Center for AIDS Research (CFAR), Biostatistics Core Focus: Sample size and power calculation, statistical analysis plan (SAP) development, data collection instruments (REDCap), and statistical analysis of interdisciplinary studies.
2019–2020	*Undergraduate Research Assistant* UNC Lineberger Comprehensive Cancer Center Focus: Differential expression analysis of tumor RNA-seq data from the TCGA database.
2021	*International Investment Deputy Department Intern* Fosun Pharmaceutical Co. Ltd.

Focus: Cross-border pharmaceutical M&A due diligence and international market analysis.

2020–2021 *Statistical Programming Intern*
Meta Clinical Technology Co. Ltd.
Focus: SAS macros, aCRF development, and statistical programming for oncology clinical trials under CDISC standards and ICH E3, E6, and E9 principles.

TECHNICAL SKILLS

Statistical computing:	R (advanced; package development), Python (proficient; PyTorch), SAS (advanced; macros)
Reproducibility & tooling:	Git/GitHub, LaTeX/Overleaf/BibLaTeX, Beamer, SLURM/HPC
Clinical data:	REDCap, CDISC standards, ICH E3/E6/E9
Languages:	English (fluent), Mandarin Chinese (native)

RESEARCH MENTORING

2026-Present *Neha Katakamshetty*, Undergraduate, Chemistry, UNC-CH;
project: multiagent and hierarchical reinforcement learning

2026-Present *Anthony Wang*, Graduate, Biostatistics, UNC-CH;
project: dynamic treatment regimes with survival outcomes

SERVICE AND LEADERSHIP

2025 *President*, UNC-CH ASA Student Chapter

2025 *Student volunteer*, UNC-CH Biostatistics Department Computing Committee

2025 *Logo design team*, UNC Gillings Center for Artificial Intelligence and Public Health

2025 *Panelist*, Student Panel: Navigating your Research Journey at UNC Biostatistics

2025 *Panel Moderator*, Career Development: Bridging the Gap Between Academia and Industry.

2024 *Vice President*, UNC-CH ASA Student Chapter

2023 *Poster judge*, UNC-CH Biostatistics Undergraduate Summer Internships (BUSI) program

2022 *Instructor volunteer*, Girls Talk Math at UNC-CH

2023-2025 *Student volunteer*, Gillings Big Buddy Program for Incoming Students (2023-2026)

On-going *Reviewer*, JRSS-C (1x), Annals of Statistics (1x), The American Statistician (1x)